

3.6 The Chain Rule

Motivation:

Suppose we want to find the first derivative of $f(x) = (2x + 1)^3$.

This idea leads to one of the most important differentiation rules: *The Chain Rule*.

Theorem (The Chain Rule): Suppose g is differentiable at x and f is differentiable at $g(x)$. Suppose that $F(x) = (f \circ g)(x) = f(g(x))$. Then, F is differentiable at x and:

$$F'(x) = f'(g(x)) \cdot g'(x)$$

In Leibniz notation, if $y = g(x)$, then we can think of f as a function of y and the chain rule becomes:

$$\frac{dF}{dx} = \frac{df}{dy} \frac{dy}{dx}$$

A partial proof of this theorem is presented in the textbook. A full proof can be found online.

Example 1: Find $F'(x)$ if $F(x) = \sqrt{x^2 + 1}$.

Example 2: Find the derivative of each of the following functions.

a) $f(x) = \sin(x^2)$

b) $g(x) = \sin^2(x)$

c) $h(x) = e^{3x}$

d) $r(x) = \frac{1}{(x^3 + 2)^5}$

We will now examine some more complicated situations which involve the chain rule.

Example 3: Suppose that f , g , and h are differentiable functions. Find a formula for the first derivative of $F(x) = f(g(h(x)))$.

Example 4: Answer each of the following:

a) Find $\frac{dy}{dx}$ if $y = e^{\sec(x^3)}$.

b) Find $\frac{dy}{dx}$ if $y = \left[\frac{e^x}{e^{2x} + 1} \right]^4$.

c) Find $\frac{dy}{dx}$ if $y = (2x+1)^5 (x^3 - x + 1)^4$.

d) Suppose that g is a differentiable function and n is some constant. Find a formula for $\frac{d}{dx} \left[(g(x))^n \right]$.

We end this section by using the chain rule to prove an important theorem.

Theorem: Suppose that b is a positive real number constant. Then, $\frac{d}{dx} (b^x) = b^x \ln(b)$.

Question to the Class: Suppose we want to find the first derivative of $y = 2^{2x}$. Are there any errors in the following two solutions?

Solution 1: By the chain rule and the above theorem: $\frac{dy}{dx} = 2^{2x} \ln(2) \cdot \frac{d}{dx}(2x) = 2^{2x} \ln(2) \cdot 2 = 2^{2x+1} \ln(2)$

Solution 2: We use some algebra, and we get $y = 2^{2x} = (2^2)^x = 4^x$. Then, $\frac{dy}{dx} = 4^x \ln(4)$.